Factorization memory in a conic ideal: the A_3, B_3, and H_3 configurations

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Working split draft — July 2026

Abstract

Restriction to a conic forgets how its marked points were paired into secants. We study the information retained after differences of matching products are divided by the conic equation. For the rank-three Coxeter configurations A_3, B_3, and H_3, the quotient has ranks 3, 6, and 10. In the balanced cases, second moments recover two complementary sheets, a cubic invariant orients them, and six incidence profiles recover the matching-table row. A modular depth quotient and arithmetic gluing describe how this factorization memory changes across characteristics.

Keywords. conic ideal; secant matching; Coxeter configuration; factorization; cubic invariant; modular representation.

2020 Mathematics Subject Classification. 05B25, 51E22, 20C20, 13A02.

1 Introduction

The companion rigidity paper isolates the Clebsch hexagon through its uncovered syndrome locus and reconstructs its projective geometry from decoding data. Here we begin with marked secant geometry and ask a complementary reconstruction question: what does common restriction to a conic forget, and what does the conic-ideal quotient remember? We answer this question across the rank-three A_3, B_3, and H_3 configurations.

The main result follows one mechanism from the switch identity to the recovery of finite matching data. The rigidity theorem motivates this question but is not a hypothesis: the marked-conic objects and quotient construction are defined independently here.

This line of research arose from the following normal-play placement game. Two players alternately adjoin a point of a finite projective plane while keeping the selected set an arc; the first player with no legal point loses. After frame reduction in an odd projective plane, a residual three-point position and the two burned directions determine a five-arc and hence a conic; adjoining an on-conic response exposes the exceptional six-arc geometry studied here. The resulting connection now runs in both directions. The small-arc equality theorem isolates the projective frame over $\mathbb{F}_5$ and the Clebsch hexagon over $\mathbb{F}_{11}$ as the only configurations of sizes four through seven whose uncovered locus is a full conic. The latter yields an icosahedral continuation complex, a certified game position, and a symmetry-compressed finite model for constructing second-player responses. Under the standard arc–MDS–AME dictionaries, the same H_3 six-arc gives an $[6, 3, 4]_{11}$ MDS code and a minimum-support stabilizer absolutely maximally entangled state on six 11-level systems, so its projective rigidity controls a concrete quantum-code input. More generally, prescribed-hole arc theory places its equality and reconstruction properties in a forward/inverse framework, while the syndrome–uncovered-point mechanism extends to projective Reed–Solomon deep holes beyond redundancy four. These are applications and continuations of the geometry, not hypotheses of the results proved here.

2 Conic matching products

Define the matching products, their common conic restriction, and the quotient obtained by dividing a difference by the conic equation. A small example should make the retained information visible before the general construction.

3 Switches and divisibility

Prove the four-endpoint switch identity and the general switch/divisibility theorem. Separate what follows formally from the conic ideal from what depends on a selected matching configuration.

4 The rank-three configurations

Introduce the A_3, B_3, and H_3 configurations with precise credit for their classical geometry. Compute the quotient ranks 3, 6, and 10, and state the complement-code calculation only at the strength needed later.

5 Balanced sheets and cubic orientation

Show how second moments recover the two balanced halves in the B_3 and H_3 cases up to interchange. Identify degree three as the first signed moment that orients the halves.

6 Six profiles and matching-row reconstruction

Derive the two $1, 4, 6$ triples from subgroup marks, construct the six incidence profiles, and explain why their rank-two image still separates all six labels. End with the exact matching-row recovery statement.

7 Modular depth and arithmetic gluing

Present the modular depth quotient, the primitive $1 : 4 : 6$ relation, and the arithmetic splitting/gluing theorem as one change-of-characteristic story. Put the relative-cubic Tate plane in an appendix unless it shortens this argument.

8 Verification architecture

Distinguish conceptual proofs, classical inputs, Lean statements, and finite certificates. State the coordinate-to-Coxeter identifications and external semantic bridges explicitly.

A The relative-cubic Tate plane

Record the two canonical constructions and the proved boundary between the relative-cubic plane and the modular depth plane.

Conclusion

State exactly which factorization data the conic ideal remembers and which pointed or global data it still forgets.